

Chapter 1

Energy Profiling of 5G User Plane Functions

1.1 Introduction

The User Plane Function (UPF) is the component of a 5G core network responsible for routing and forwarding GTP-U encapsulated user traffic between the Radio Access Network (RAN) and the Data Network (DN). Its power consumption is therefore directly coupled to network load, and understanding this relationship is a prerequisite for energy-aware network management and for meeting increasingly stringent sustainability targets in mobile telecommunications.

This chapter presents a systematic empirical study of two open-source UPF implementations under controlled laboratory conditions. The first implementation, SD-CORE DPDK, is based on the SD-Core project and leverages the Data Plane Development Kit (DPDK) poll-mode I/O driver. The second, OAI-UPF, is the user-plane component of OpenAirInterface and uses a conventional interrupt-driven Linux I/O model. Traffic was generated with a Keysight LoadCore emulator, and power was measured at the host level using the Scaphandre process-level power monitor with 1 s sampling resolution.

The remainder of this chapter is organised as follows. Section 1.2 describes the measurement methodology and dataset. Section 1.3 characterises each variant across performance and energy dimensions. Section 1.4 investigates the saturation behaviour of OAI-UPF in depth. Section 1.5 analyses the Specific Energy Consumption metric and its interpretation for poll-mode versus interrupt-driven drivers. Section 1.6 reports feature correlations relevant to predictive energy modelling. Section 1.7 discusses implications for network operation and modelling.

1.2 Measurement Methodology

1.2.1 Testbed Configuration

Both UPF variants were deployed on identical server hardware. LoadCore injected downlink UDP flows with a fixed packet size of 1250 B at progressively increasing offered loads. Each load step was sustained for a fixed duration to allow steady-state measurements; the raw time-series data were then merged with per-second power readings from Scaphandre.

1.2.2 Dataset

After preprocessing, the merged dataset comprises 9556 one-second measurement samples across 110 OAI runs and an equivalent number of DPDK runs. Following feature engineering (removal of zero-variance and near-duplicate columns, unit conversions, target computation), the analysis dataset contains 8781 samples and 44 columns, split evenly between the two variants (4388 DPDK, 4393 OAI).

1.2.3 Metrics

The following primary metrics are reported throughout this chapter:

Power (W) Total host power consumption in watts, as reported by Scaphandre (`power_watts`).

Net power (W) Incremental power above the idle baseline: $P_{\text{net}} = P_{\text{total}} - P_{\text{idle}}$.

SEC_{total} Specific Energy Consumption based on total power: $\text{SEC}_{\text{total}} = P_{\text{total}}/\lambda$, where λ is throughput in Gbps.

SEC_{net} Marginal energy cost: $\text{SEC}_{\text{net}} = P_{\text{net}}/\lambda$.

Packet loss Count of GTP-U packets lost per one-second interval at the DN-side receiver.

Downlink delay Weighted mean one-way packet delay on the downlink path (μs).

1.3 Variant Profiles

1.3.1 Overview

Table 1.1 summarises the key performance and energy characteristics of both variants across all measurement conditions. Figure 1.1 illustrates the fundamental difference in the power–throughput relationship.

Table 1.1: UPF variant profiles: key performance and energy metrics.

Metric	SD-Core DPDK	OAI-UPF
<i>Power</i>		
Total power range (W)	0.807–0.859	0.000–4.497
Mean power (W)	0.821	1.435
Standard deviation (W)	0.005	1.527
Dynamic range (W)	0.052	4.497
Net power mean (W)	0.001	1.419
<i>CPU</i>		
Mean CPU (%)	1.064	1.549
Max CPU (%)	1.079	4.554
Idle CPU (%)	1.064	0.018
<i>Throughput</i>		
Maximum (Gbps)	5.06	2.33
Mean (Gbps)	0.788	0.528
<i>Reliability</i>		
Packet loss (sample rows, %)	0.0	32.7
<i>Latency</i>		
DL delay mean (μ s)	311	5031
DL delay max (μ s)	24978	25000
RAN jitter mean (μ s)	1.3	1428
<i>Energy efficiency at >1 Gbps</i>		
SEC _{total} median (J/Gb)	0.27	1.61
SEC _{net} median (J/Gb)	0.002	1.61
L2/L3 overhead ratio (mean)	2.11	1.86

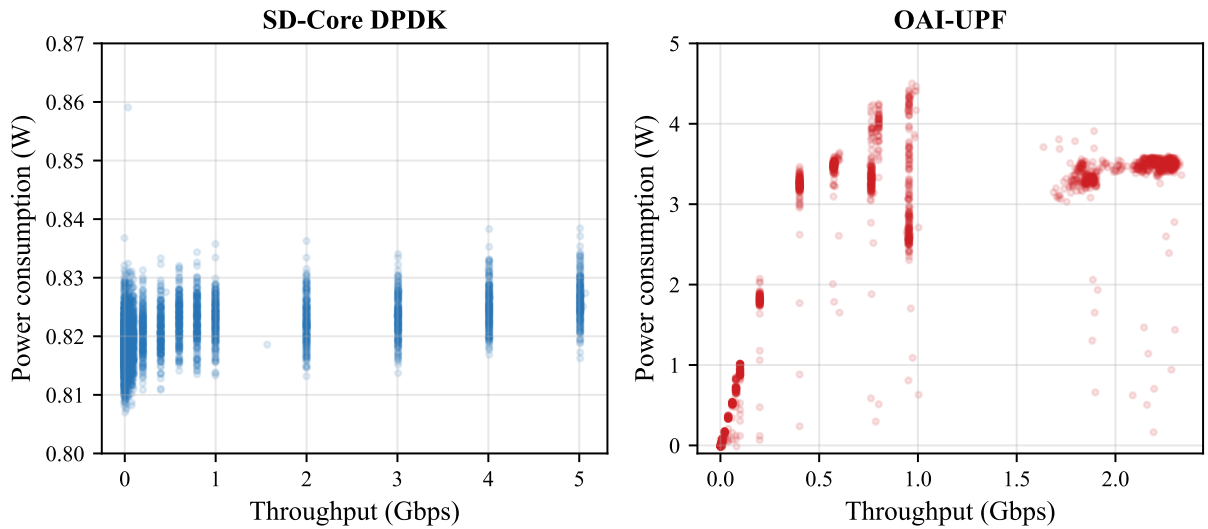


Figure 1.1: Power consumption versus throughput for each UPF variant. SD-Core DPDK (left) exhibits a near-constant power profile due to continuous CPU polling. OAI-UPF (right) shows a strong load-proportional relationship but saturates above 0.5 Gbps.

1.3.2 SD-Core DPDK

The DPDK data-plane driver operates in poll mode: a dedicated CPU thread continuously spins on the receive queue, eliminating interrupt overhead at the cost of occupying the core regardless of traffic. This architectural choice manifests clearly in the measurements.

Power consumption remains almost constant at 0.821 ± 0.005 W across the entire load range (0–5.06 Gbps), yielding a dynamic range of only 52 mW. CPU utilisation is similarly invariant at 1.064 ± 0.004 %, and the Pearson correlation between throughput and CPU is $r = 0.016$, confirming that the polling loop consumes a fixed share of the core irrespective of packet rate.

Despite its flat power profile, SD-CORE DPDK achieves the highest throughput of both variants (5.06 Gbps) with zero packet loss across all 4388 measurement samples. Downlink delay is low (mean $311 \mu\text{s}$) and RAN jitter is negligible (mean $1.3 \mu\text{s}$), making it the preferable choice for latency-sensitive traffic.

1.3.3 OAI-UPF

OAI-UPF uses the standard Linux socket and eBPF/XDP interrupt path. Power scales strongly with offered load: $r(\text{throughput}, \text{power}) = 0.81$ (compared with 0.46 for DPDK), and the system transitions from near-zero power (0.034 W) at idle to a plateau of approximately 3.4 W once saturated. CPU utilisation follows the same trend ($r = 0.76$), rising from 0 % at idle to 4.6 % at peak.

However, OAI-UPF saturates at approximately 0.5 Gbps (see Section 1.4), beyond which virtually all packets are dropped. Delay is consistently higher than DPDK (mean 5 ms) and RAN jitter degrades catastrophically above saturation (mean $1428 \mu\text{s}$, maximum 199 ms).

Figure 1.2 illustrates the load-stratified comparison of both variants across the four key metrics.

1.4 OAI-UPF Saturation Analysis

1.4.1 Saturation Onset

To characterise the saturation regime with fine granularity, measurements were binned into 50 Mbps throughput intervals. Figure 1.3 shows the transition across four dimensions: loss incidence, loss magnitude, delay, and the power/CPU trajectory.

The first bin where more than 5 % of samples exhibit packet loss occurs at ≈ 0.17 Gbps, indicating the onset of saturation. Loss increases steeply, reaching 50 % at ≈ 0.57 Gbps, beyond which essentially all samples report loss. From this point, OAI-UPF enters a fully

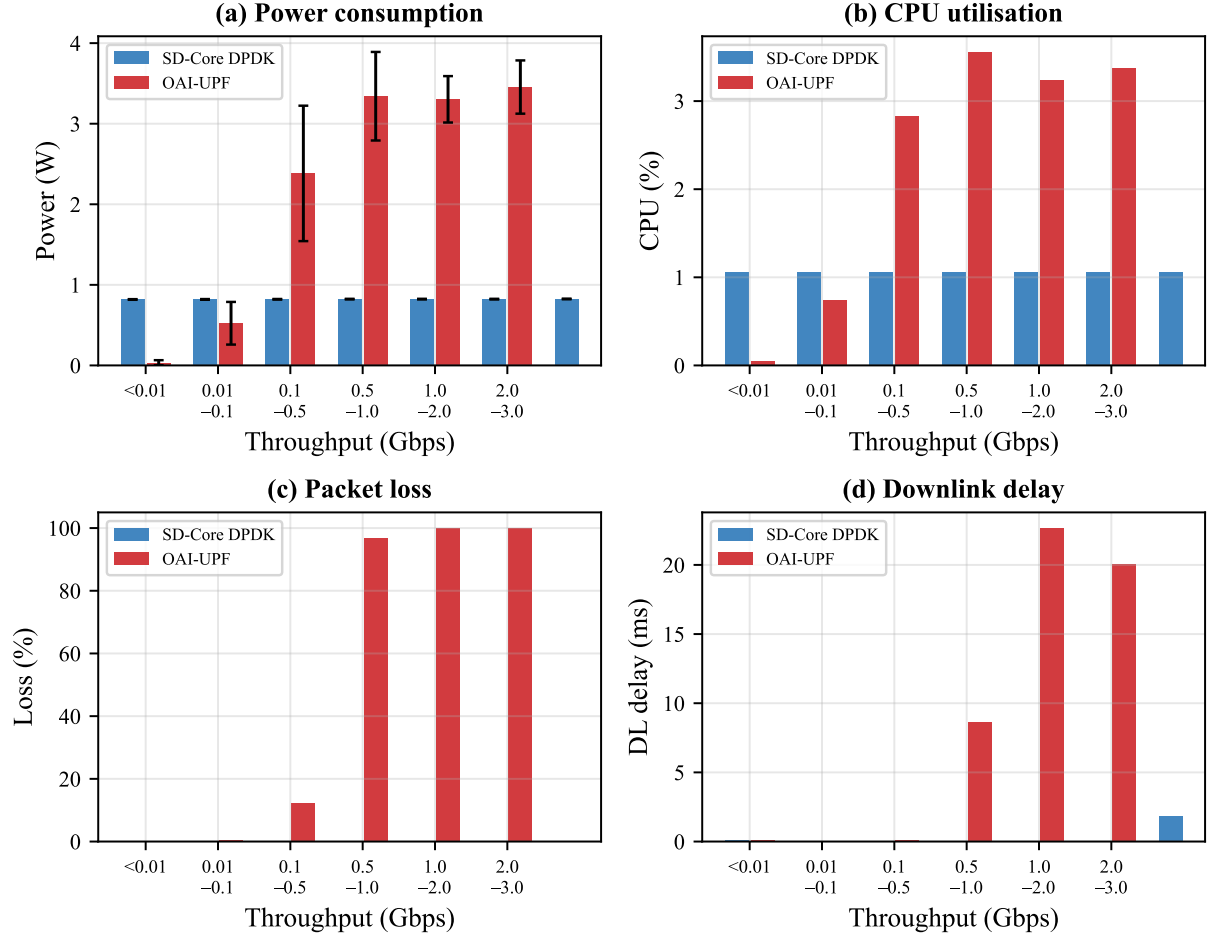


Figure 1.2: Load-stratified variant profiles. Throughput bins on the x-axis (Gbps). (a) Power consumption; (b) CPU utilisation; (c) packet loss incidence; (d) mean downlink delay. Error bars in (a) denote ± 1 standard deviation. DPDK maintains flat power and zero loss at all loads; OAI scales with load but saturates above 0.5 Gbps.

saturated regime: throughput measurements reflect the offered load injected by LoadCore, not the effective forwarding rate, and delay jumps to the measurement ceiling of 25 ms.

OAI-UPF Saturation Transition

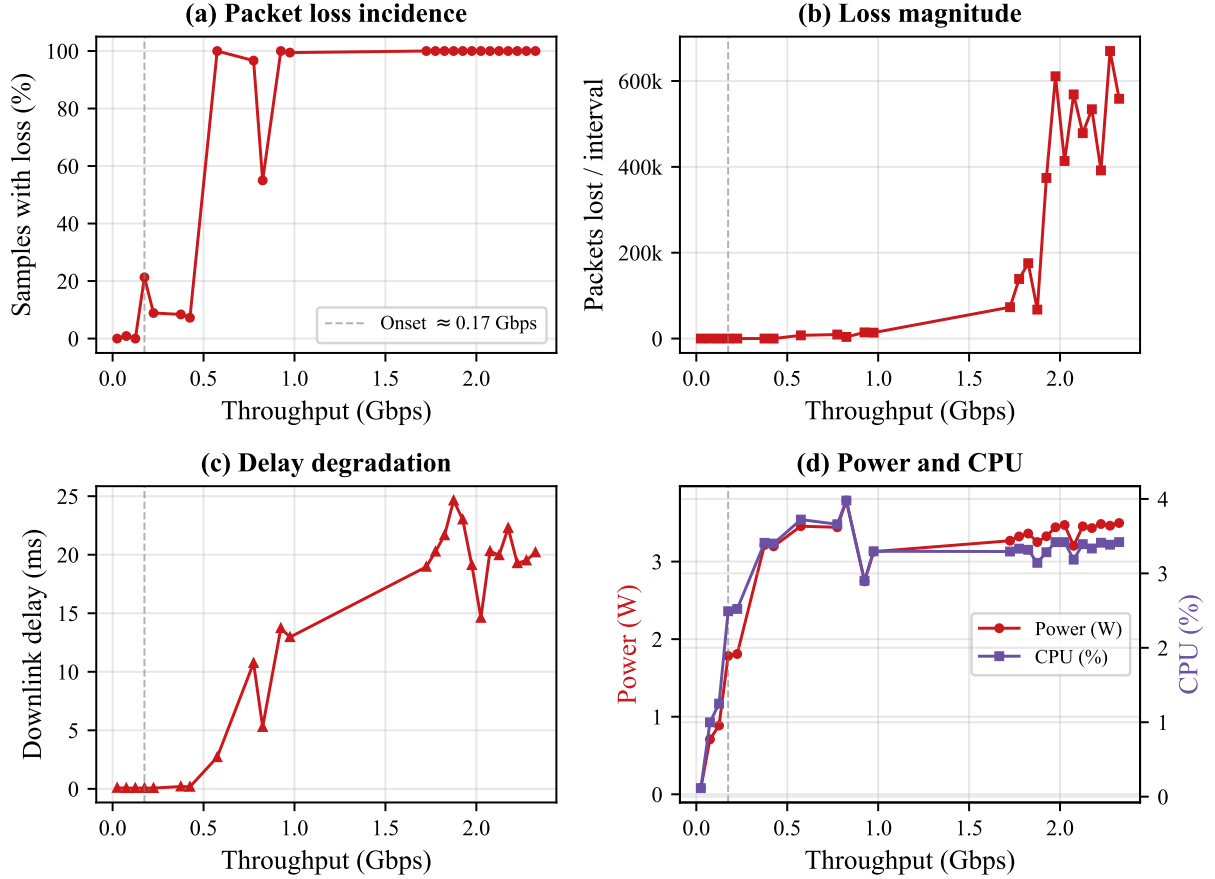


Figure 1.3: OAI-UPF saturation transition in 50 Mbps throughput bins. The dashed vertical line marks the saturation onset at ≈ 0.17 Gbps. (a) Fraction of samples with packet loss; (b) mean number of packets lost per interval (when loss occurs); (c) mean downlink delay; (d) mean power and CPU utilisation (dual axis).

1.4.2 Power Behaviour at Saturation

Despite losing packets, OAI-UPF continues to consume power proportionally to the offered load up to ≈ 0.5 Gbps, after which power plateaus at approximately 3.4 W (Figure 1.3d). This indicates that the CPU is fully committed to packet processing at saturation; additional offered load does not increase power but increases loss instead.

The behaviour is summarised as follows:

- Pre-saturation (< 0.3 Gbps): mean power = 0.338 W, zero loss.
- Transition (0.3–0.57 Gbps): power rises sharply, loss begins.
- Post-saturation (> 0.5 Gbps): mean power ≈ 3.4 W, near-total loss.

1.4.3 Per-Run Characterisation

Figure 1.4 aggregates the loss and power behaviour at the granularity of individual measurement runs. Of the 110 OAI runs, 61 (55 %) completed without any packet loss; these runs had a maximum offered throughput of at most 0.1 Gbps. Every run exceeding 0.06 Gbps maximum offered load incurred at least some packet loss, establishing the *practical safe operating boundary* of OAI-UPF at approximately 100 Mbps.

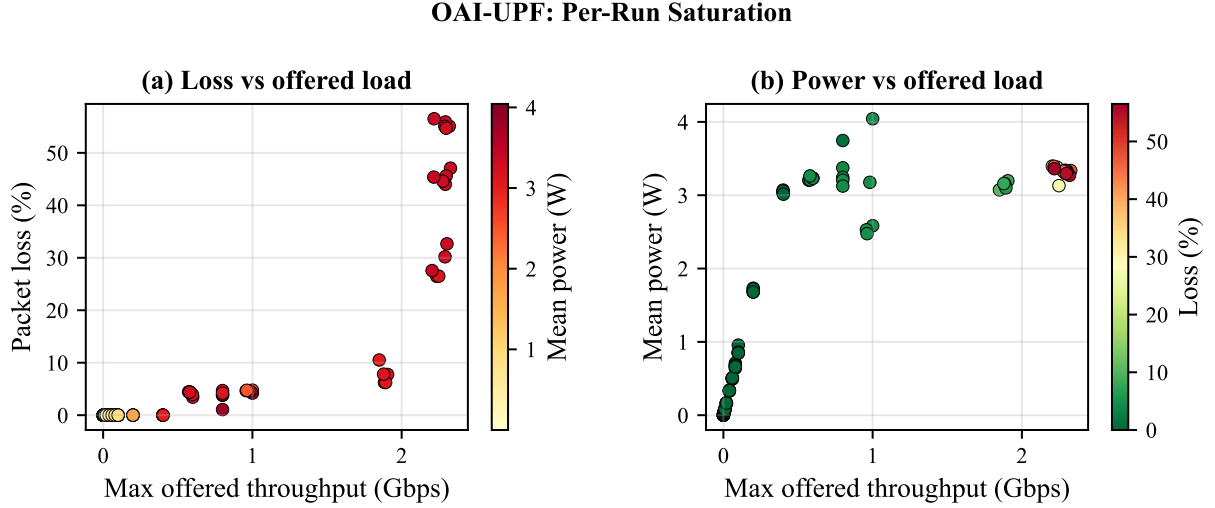


Figure 1.4: OAI-UPF per-run saturation summary. Each point represents one measurement run. (a) Overall packet loss versus maximum offered throughput, coloured by mean power; (b) mean power versus maximum offered throughput, coloured by packet loss fraction.

1.5 Specific Energy Consumption Analysis

1.5.1 Definitions

Specific Energy Consumption (SEC) is defined as the ratio of power consumed to useful throughput delivered. Two variants are considered:

$$\text{SEC}_{\text{total}} = \frac{P_{\text{total}}}{\lambda} \quad [J/Gb] \quad (1.1)$$

$$\text{SEC}_{\text{net}} = \frac{P_{\text{net}}}{\lambda} = \frac{P_{\text{total}} - P_{\text{idle}}}{\lambda} \quad [J/Gb] \quad (1.2)$$

1.5.2 The Poll-Mode Measurement Paradox

Equation (1.2) was introduced in the literature to capture the *marginal* energy cost of transmitting one additional gigabit, isolating the load-dependent component from the fixed idle overhead. While this definition is appropriate for interrupt-driven systems (where $P_{\text{idle}} \ll P_{\text{total}}$), it produces a misleading result for poll-mode drivers.

In SD-CORE DPDK, the CPU polls continuously regardless of load, so $P_{\text{idle}} \approx P_{\text{total}} \approx 0.821 \text{ W}$ at all times. Consequently, $P_{\text{net}} = P_{\text{total}} - P_{\text{idle}} \approx 0$, and SEC_{net} approaches zero as throughput increases, superficially suggesting near-perfect efficiency. In reality, SD-CORE DPDK consumes 0.82 W regardless of whether it forwards 0 or 5 Gbps.

Table 1.2 and Figure 1.5 quantify this discrepancy at high throughput ($> 1 \text{ Gbps}$).

Table 1.2: Specific Energy Consumption at high throughput ($> 1 \text{ Gbps}$): median values.

Metric	SD-Core DPDK	OAI-UPF
Total power at $>1 \text{ Gbps}$ (W)	0.824	3.478
$\text{SEC}_{\text{total}}$ (J/Gb)	0.274	1.615
SEC_{net} (J/Gb)	0.002	1.607

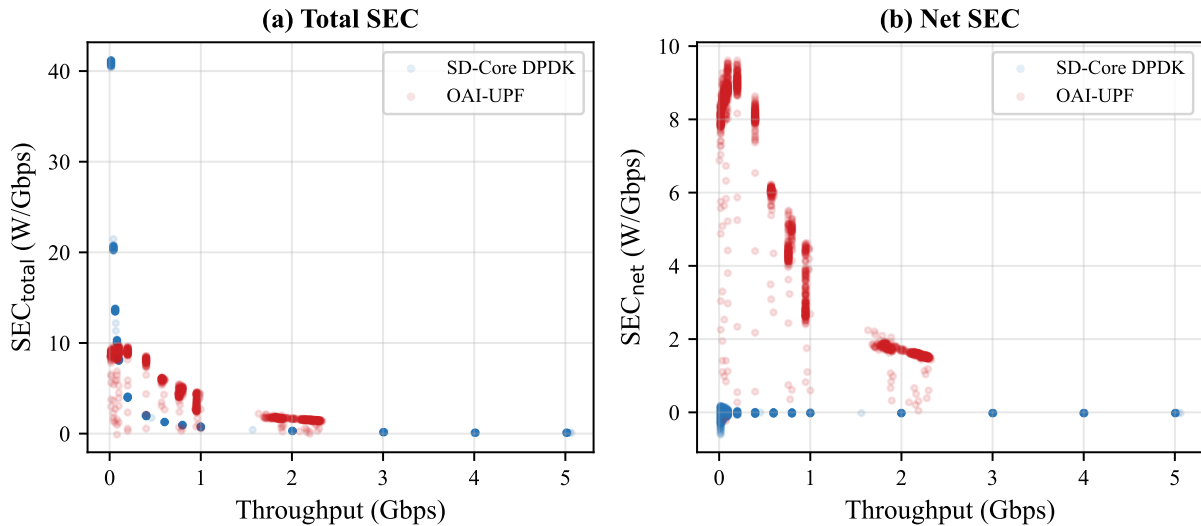


Figure 1.5: Specific Energy Consumption versus throughput. (a) $\text{SEC}_{\text{total}}$: DPDK appears efficient due to low power denominator relative to flat numerator. (b) SEC_{net} : DPDK’s SEC_{net} collapses to near zero because its incremental power $P_{\text{net}} \approx 0$. Both metrics understate the operational energy cost of running a poll-mode UPF. Extreme low-throughput outliers are clipped to the 98th percentile for clarity.

1.5.3 Implications for Energy Metric Selection

For a fair cross-variant comparison, $\text{SEC}_{\text{total}}$ is the more honest metric: it reflects the full operational cost of sustaining throughput λ . At $> 1 \text{ Gbps}$, DPDK achieves 0.27 J/Gb and OAI achieves 1.61 J/Gb —a $6\times$ difference in total power per bit. However, total power consumption must also be considered in absolute terms: at idle, DPDK draws 0.82 W whereas OAI draws less than 0.04 W . The preferable variant therefore depends on the traffic load distribution in deployment.

1.6 Feature Correlation Analysis

1.6.1 Correlates of Power Consumption

Table 1.3 lists the ten strongest Pearson correlations with `power_watts` in the pooled dataset (excluding features derived directly from the target).

Table 1.3: Top feature correlations with power consumption (pooled dataset). Rolling features are marked † to indicate data-leakage risk when used as predictors.

Feature	Pearson r	Note
<code>power_watts_roll_max_30s</code> †	+0.998	Derived from target
<code>cpu_pct</code>	+0.991	Strong legitimate predictor
<code>power_watts_roll_mean_30s</code> †	+0.982	Derived from target
<code>cpu_per_watt</code>	−0.936	Derived from target
<code>dl_delay_power_product</code>	+0.797	Interaction feature
<code>high_delay_frac</code>	+0.768	QoS indicator
<code>weighted_mean_delay_us</code>	+0.735	Latency feature
<code>ul_delay_power_product</code>	+0.614	Interaction feature
<code>l2l3_overhead_ratio</code>	−0.561	Encapsulation efficiency
<code>packets_lost_delta</code>	+0.535	Loss indicator

Figure 1.6 shows the correlation structure separately for each variant. The two variants exhibit markedly different correlation patterns: in SD-CORE DPDK, power is weakly correlated with all traffic metrics ($r < 0.5$) because the power signal is nearly flat; in OAI-UPF, the throughput–power correlation is $r = 0.81$ and CPU–power is $r = 0.92$, making power highly predictable from traffic observables.

1.6.2 Temporal Dynamics

Figure 1.7 shows representative power and throughput traces within a single high-load run for each variant. SD-CORE DPDK exhibits essentially no power variation despite large throughput swings; OAI-UPF tracks load closely until saturation, then power plateaus while throughput (offered load) continues to increase.

1.7 Discussion and Implications

1.7.1 Architectural Trade-offs

The measurements reveal a fundamental trade-off between the two I/O paradigms:

Poll mode (DPDK) delivers superior throughput (5.06 vs 2.33 Gbps), zero packet loss, and deterministically low latency, at the cost of a fixed power floor of 0.82 W regard-

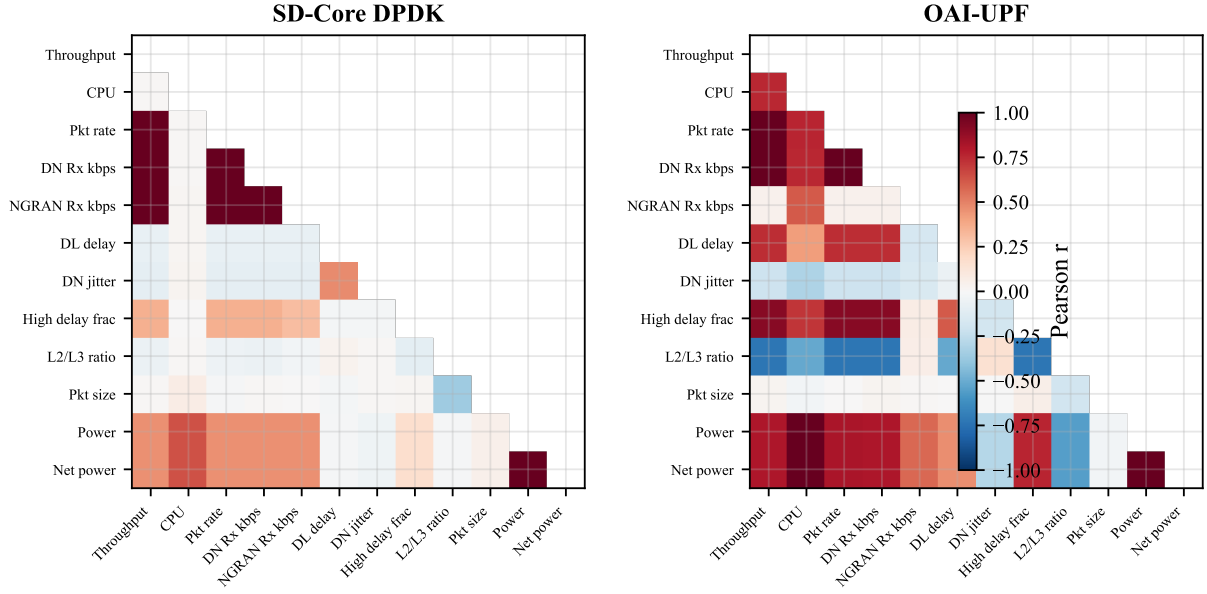


Figure 1.6: Pearson correlation heatmaps for a subset of key features, computed separately for each variant. The DPDK heatmap (left) shows uniformly low correlations with power, consistent with the near-constant power profile. The OAI heatmap (right) reveals strong positive correlations between throughput, CPU, and power.

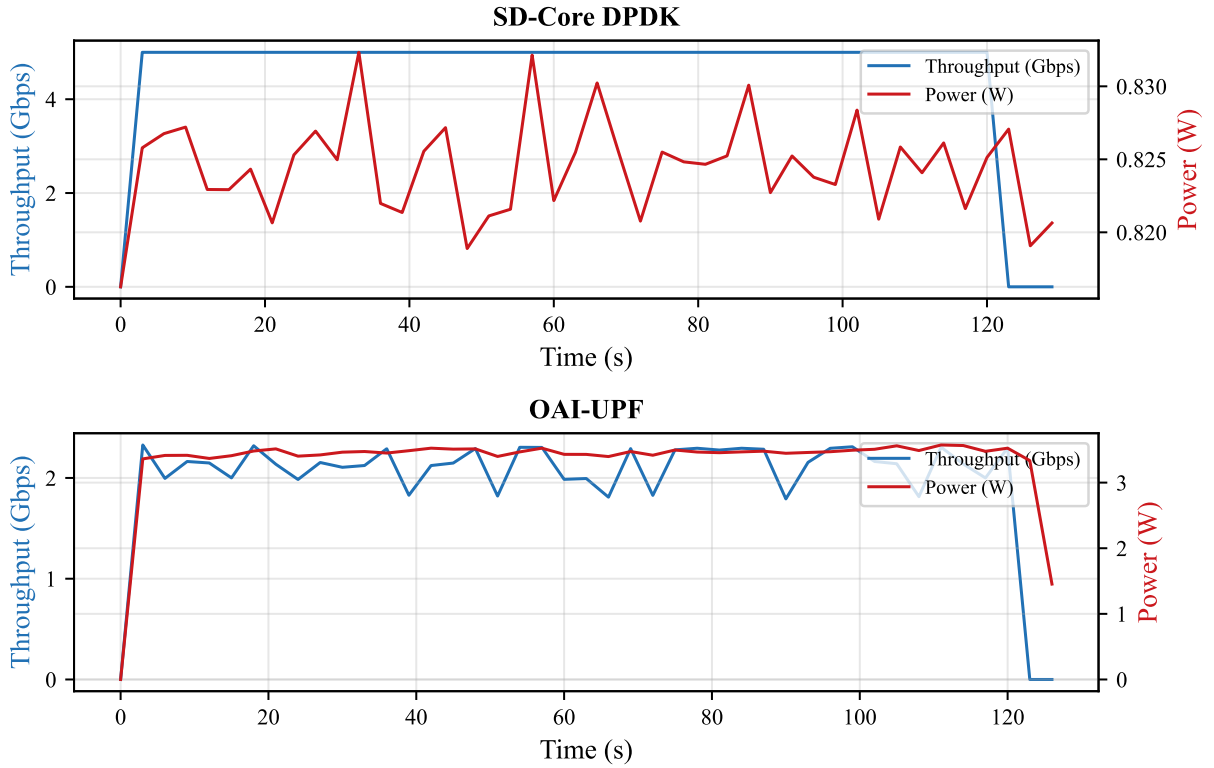


Figure 1.7: Temporal power and throughput traces for representative high-load runs. SD-Core DPDK (top): power remains flat despite throughput variations. OAI-UPF (bottom): power rises with load until the saturation plateau.

less of load. This is appropriate for high-throughput, latency-sensitive deployments where the UPF is expected to be consistently busy.

Interrupt mode (OAI) scales power proportionally to load, drawing near-zero power at idle, but saturates at ≈ 0.5 Gbps and incurs severe delay and loss in the saturated regime. It is preferable only in low-to-medium load scenarios where energy-proportionality is prioritised over peak throughput.

1.7.2 Safe Operating Regions

Based on the per-run saturation analysis (Section 1.4, Figure 1.4), the practical safe operating boundary for OAI-UPF is $\lambda \leq 0.10$ Gbps. Above this threshold, packet loss becomes probable; above 0.57 Gbps, it is near-certain. No such constraint applies to SD-CORE DPDK within the measurement range.

1.7.3 Implications for Predictive Energy Modelling

The near-constant power of SD-CORE DPDK presents a challenge for regression-based energy models: the target variable has a dynamic range of only 52 mW, approaching the noise floor of Scaphandre. Models trained on DPDK data are expected to predict 0.82 W for almost any input; the information content of the traffic features is minimal. Per-variant models are therefore strongly recommended over a single pooled model.

For OAI-UPF, the strong linear relationships ($r > 0.8$) between throughput, CPU, and power suggest that simple linear or low-complexity tree models may suffice in the unsaturated regime. Training should be restricted to the safe operating region ($\lambda < 0.5$ Gbps) unless the modelling objective explicitly includes the saturated regime.

The rolling power features (`power_watts_roll_mean_30s`, `power_watts_roll_max_30s`) achieve very high correlations with the power target ($r > 0.98$) and must be excluded as predictors when the modelling goal is to predict power from non-power observables.

1.8 Summary

This chapter characterised two 5G UPF implementations under controlled load conditions. The principal findings are:

1. **DPDK is power-constant:** total power varies by only 52 mW over the full 0–5 Gbps load range due to the CPU polling loop. This makes power prediction from traffic features statistically intractable.
2. **OAI is power-proportional but capacity-limited:** power scales linearly from ≈ 0 to ≈ 3.4 W, but packet loss becomes near-universal above 0.57 Gbps. The practical operating boundary is ≈ 0.1 Gbps.

3. **The SEC measurement paradox:** the SEC_{net} metric misleadingly reports near-zero energy efficiency for DPDK because the idle power is almost equal to the load power. Total SEC ($\text{SEC}_{\text{total}}$) is the more operationally meaningful metric.
4. **CPU is the strongest legitimate predictor of power** ($r = 0.99$ pooled), followed by delay-related features. Rolling power features are informative but constitute data leakage when used as predictors of power.
5. **Separate per-variant models are necessary:** the two variants occupy distinct regimes with different predictive structures and cannot be modelled jointly without a variant indicator.